(4d) Determine yield strength for 0.1% [illegible]

$$\frac{\delta\sigma}{\delta\varepsilon} = \frac{1273.24\,psi}{0.00033} = \underline{\underline{3858303.03\,psi}}$$

Yield strength = 11000 psi (See attached graph for visualization)

(4e) $\underline{\underline{\text{Ultimate strength} = 11204.51\,psi}}$

(4f) $E_0 = 3858303.03\,psi$ (Calculated for offset method in 4d)

(4g) $E_{sec,0.004} = \frac{11077.18\,psi}{0.004} = \underline{\underline{2769295\,psi}}$

(6) Aluminium block 5cm thick that is 75cm by 75cm. This is a 100000kN lateral load and G = 27GPa, what is the shear displacement, in mm?

Assuming linear elastic behavior

$\gamma = G\varepsilon$

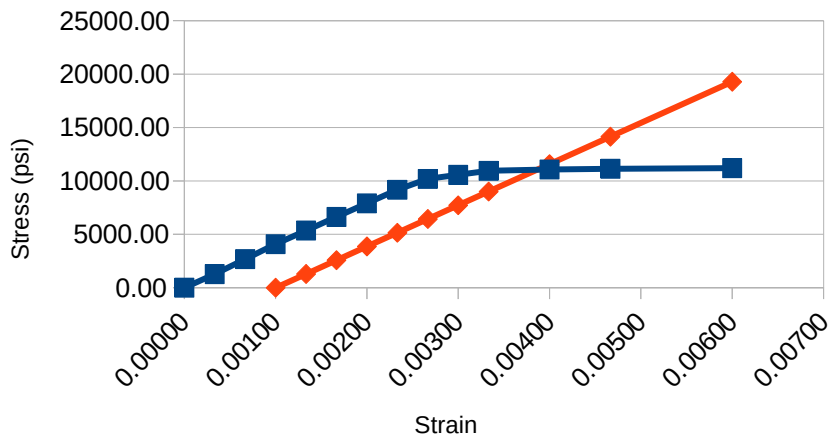
$\gamma = G\frac{V}{A}$

$\gamma = 27GPa \cdot \frac{100000N}{0.75m} = .36$

$\underline{\underline{d = 750mm \cdot .36 = 270mm}}$

Load (lb)	Elongation (in)	Stress (psi)	Strain	Strain Energy (lb-ft)	Strain Energy Density (psi)
0	0.000	0.00	0.00000	0.00000	0.00000
40	0.001	1273.24	0.00033	0.02000	0.21221
84	0.002	2673.80	0.00067	0.08400	0.89127
128	0.003	4074.37	0.00100	0.19200	2.03718
168	0.004	5347.61	0.00133	0.33600	3.56507
208	0.005	6620.85	0.00167	0.52000	5.51737
248	0.006	7894.09	0.00200	0.74400	7.89409
288	0.007	9167.32	0.00233		
320	0.008	10185.92	0.00267		
332	0.009	10567.89	0.00300		
344	0.010	10949.86	0.00333		
348	0.012	11077.18	0.00400		
350	0.014	11140.85	0.00467		
352	0.018	11204.51	0.00600		

(c) Stress-Strain Graph



Formula for 0.1% Offset Yield Strength	Ultimate Strength
-3858.30303	11204.50799
-2572.20202	
-1286.10101	
0.00000	
1286.10101	
2572.20202	
3858.30303	
5144.40404	
6430.50505	
7716.60606	
9002.70707	
11574.90909	
14147.11111	
19291.51515	